

DEFINICIJE 1

Navadna diferencialna enačba 1. reda $\dot{x} = F(x, t)$

Skalarne linearne enačbe 1. reda $\dot{x} = f(t)x + g(t)$

Integralne krivulje $\dot{y}(t) = \vec{f}(y(t))$

Eksistenčni izrek $x_r(0) = x_0$ $\dot{x} = F(t, x)$

• F Lipschitzova na $\vec{x} \geq L$ • $F \leq M$ na $[t_0 - a, t_0 + a] \times B_0$

Potem rešitev obstaja in je enolična na

$[t_0 - \alpha, t_0 + \alpha] \rightarrow B_0$, kjer $\alpha \leq \min \left\{ \frac{1}{2M}, \frac{1}{L} \right\}$

Prvi integral u za $y' = F(x, y)$, če za $\forall$ rešitev $y(x)$
velja $u(x, y(x)) \equiv C$

Pfaffova enačba $P(x, y)dx + Q(x, y)dy = 0$

Integrirajoči množitelj $\lambda(x, y)$, da je $\lambda \cdot F$ potencialno

Lagrange d'Alembertova formula $y = x \cdot \varphi(y') + \psi(y')$

Clairnotova enačba $y = x y' + \psi(y')$

Ovojnica družine krivulj $F(x, y; c) \equiv 0$ je $c \mapsto (x(c), y(c))$
za katero velja $\forall c. (x'(c), y'(c)) \perp (\dot{x}(c, t), \dot{y}(c, t))$

DEFINICIJE 2

Picardova preslikava $\mathcal{A}(\vec{x}) = \vec{x}_0 + \int_0^t F(\tau, x(\tau)) d\tau$

Peznov izrek • $F(t, \vec{x}) : [t_0 - a, t_0 + a] \times B_{b, x_0} \longrightarrow \mathbb{R}^n$ zvezna

Potem $\exists x$ da obstaja vsaj ena rešitev $\vec{y} = F(t, \vec{y})$, $\vec{y}(t_0) = \vec{x}_0$

Tok vektorskega polja $\Phi_t(\vec{x}_0) = \vec{y}(t, x_0)$ za $\vec{y}$ rešitev problema $\dot{\vec{y}} = F(t, \vec{y})$ $y(t_0) = x_0$

Infinitesimalni generator $\{\Phi_t\}$ je polje, da velja

$$\frac{d}{dt} \Big|_{t=0} \Phi_t(x) = F(x)$$

Kompletno polje F , če za $\forall x_0 \in \Omega$ je rešitev problema $\dot{\vec{x}} = F(\vec{x})$, $\vec{x}(0) = \vec{x}_0$ definirana za $\forall t \in \mathbb{R}$

Hamiltonski sistem $\Omega \subseteq \mathbb{R}^{2n}$ $\Omega = \{(q_1, \dots, q_n, p_1, \dots, p_n)\}$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Liouvillova enačba • Φ_t tok vekt. polja F na $\Omega \subseteq \mathbb{R}^{2n}$

$$\frac{d}{dt} \det(J\Phi_t(\vec{a})) = \operatorname{div}(F(\Phi_t(\vec{a}))) \cdot \det(J\Phi_t(\vec{a}))$$

$$\text{Oziram } \det(J\Phi(\vec{a})) = e^{\int \operatorname{div}(F(\tau, \vec{a})) d\tau}$$

Hamiltonsko polje $X_H = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \nabla H$

Nepopolni eliptični integral 1. vrste $F(z, k) = \int_0^z \frac{d\tau}{\sqrt{1 - k^2 \sin^2(\tau)}}$

Jacobijeva amplituda $\operatorname{am}(z, k) = F^{-1}(z, k) \nearrow$

Poissonov oklepaj $\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right) = X_f \cdot \nabla g$

Liejev produkt vektorskih polj

• antisimetričnost $[F, G] = -[G, F]$

• Jacobijeva identiteta

$$[E, [F, G]] + [G, [E, F]] + [F, [G, E]] = 0$$

Simetrija Hamiltonskega sistema (H, Ω) je difeomorfizem

$S: \Omega \longrightarrow \Omega$, da γ rešitev za (H, Ω) $\gamma(t) = (p(t), q(t))$

$\gamma(0) = x_0$, potem je $S(\gamma(t))$ rešitev za (H, Ω) z

začetnim pogojem $S(\gamma(0)) = S(x_0)$

DEFINICIJE 3

$$\dot{\vec{x}} = A(t) \vec{x}$$

Fundamentalna matrika $\Phi_{t_0, t}$ sistema $\vec{x} = A(t) \vec{x}$ je
 $\Phi = [\vec{x}_1 \dots \vec{x}_n]$ kjer so $\vec{x}_i$ baza rešitev. $\vec{x}_i(t_0) = \vec{e}_i$

Ekspenzijacija matrike $e^A = I + \sum_{n=1}^{\infty} \frac{1}{n!} A^n$

BCH formula $e^A e^B = e^C \leadsto C = \ln(e^A e^B)$

Dysonova vrsta $\Phi_t = I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1) A(\tau_2) \dots A(\tau_n) d\vec{\tau}$

Liouvillova formula $\det(\Phi_t) = \exp\left(\int_{t_0}^t \text{tr}(A(\tau)) d\tau\right) \det(\Phi_{t_0})$
 $\det(\Phi_t) = \det(\Phi_{t_0}) e^{\int_{t_0}^t \text{tr}(A(\tau)) d\tau}$

Determinanta wronskega $W(t) = \det(\Phi_t)$

DEFINICIJE 4

Lagrangeev funkcional $\mathcal{L}(y) = \int_a^b L(x, y(x), y'(x)) dx$

Šibki odvod (smerni, Gâteauxov) $P: U \rightarrow V$

$$U_{u_0}^{\circ}(v) = \lim_{t \rightarrow 0} \frac{1}{t}(P(u_0 + tv) - P(u_0)) = \left. \frac{d}{dt} \right|_{t=0} P(u_0 + tv)$$

Krepi odvod (Frechetov) $P: U \rightarrow V$, $A: U \rightarrow V$ linearna, omejena. $P(u_0 + v) = P(u_0) + A(v) + o(u_0, v)$; $\lim_{\|v\| \rightarrow 0} \frac{\|o\|}{\|v\|} = 0$

Dopustna funkcija $v \in \{u \in U, u(a) = A, u(b) = B\}$

Dopustna variacija $v \in \{u \in U, u(a) = 0, u(b) = 0\}$

Euler-Lagrangeova enačba $\frac{\partial L}{\partial y}(x, y(x), y'(x)) - \frac{d}{dx} \frac{\partial L}{\partial y'}(\dots) = 0$

Vezani ekstrem: Pogoji $\Phi(u) = l \in \mathbb{R}$, ekstrem $\mathcal{L}$

KAJ MORAM ZNATI

- Kako vemo, da je rešitev sistema globalno definirana
- Dysonova vrsta
- Liouvillova formula
- Vezen: ekstremi
- Euler-Lagrangova enačba
- Tangentni prostor je enake jedru vezi
- Eksistenčni izrek
- Variacijski račun
- odvod L
- dopustne variacije
- tok in lastnosti
- primeri za eksistenčni izrek
- Fundamentalne metrike

Eksistenčni izrek

• $F(t, \vec{x}) : [t_0 - a, t_0 + a] \times B_b \rightarrow \mathbb{R}$ zvezna, Lipschitzova na drugi spremenljivki s konstanto L .

• $\dot{\vec{y}} = F(t, \vec{y}) \quad \vec{y}(t_0) = \vec{x}_0 \in B_{\frac{b}{2}}$

Potem obstaja natanko ena rešitev problema

$$\vec{y} : [t_0 - \alpha, t_0 + \alpha] \rightarrow B_b \quad \alpha \leq \min \left\{ \frac{b}{2M}, \frac{1}{L} \right\}$$

$$M = \max \{ F(t, \vec{x}) ; (t, \vec{x}) \in [t_0 - a, t_0 + a] \times B_b \}$$

Dokaz:

$$\vec{x}(t) = \vec{x}_0 + \int_{t_0}^t F(\tau, \vec{x}(\tau)) d\tau$$

$$\mathcal{A}(\vec{x}) = \vec{x}_0 + \int_{t_0}^t F(\tau, \vec{x}(\tau)) d\tau \quad \mathcal{A} : V \rightarrow V$$

$\vec{x}$ je rešitev $\Leftrightarrow \vec{x}(t) = \mathcal{A}(\vec{x}(t)) \Leftrightarrow \vec{x}$ je fiksna točka.

Take točke obstaja, ko je $\mathcal{A}$ skrajitev

$$P_{a,b} = \{ (t, \vec{x}) \in \mathbb{R} \times \mathbb{R}^n ; |t - t_0| < a, \|\vec{x} - \vec{x}_0\| \leq b \}$$

$$E = \{ \vec{y} : \mathcal{I} \rightarrow \mathbb{R}^n ; \vec{y}(t_0, \vec{x}) = \vec{x} ; \|\vec{y}(t, \vec{x}) - \vec{x}_0\| \leq b \quad \forall (t, \vec{x}) \in P_{a,b/2} \}$$

$$\mathcal{I} = [t_0 - \alpha, t_0 + \alpha] \times B_{\frac{b}{2}}$$

$$\mathcal{A}(E) \subseteq E$$

$$1) \mathcal{A}(\vec{y}(t_0, \vec{x})) = \vec{x} \quad \mathcal{A}(\vec{y}) - \vec{x} = \int_{t_0}^t F(\tau, \vec{y}(\tau, \vec{x})) d\tau = 0$$

$$2) \|\mathcal{A}(\vec{y}(t, \vec{x})) - \vec{x}_0\|_\infty \leq \|\vec{x} - \vec{x}_0\| + \int_{t_0}^t \|F(\tau, \vec{y}(\tau, \vec{x}))\| d\tau \leq \frac{b}{2} + \alpha M \leq b \quad \text{za} \quad \alpha \leq \frac{b}{2M}$$

Ali je $\mathcal{A}$ na E skrajitev?

$$\begin{aligned} \|\mathcal{A}(\vec{y}_1) - \mathcal{A}(\vec{y}_2)\| (t, \vec{x}) &= \left\| \int_{t_0}^t F(\tau, \vec{y}_1) - F(\tau, \vec{y}_2) d\tau \right\|_\infty \leq \\ &\leq \left\| \int_{t_0}^t L \|\vec{y}_1(\tau, \vec{x}) - \vec{y}_2(\tau, \vec{x})\| d\tau \right\|_\infty \leq \alpha L \|\vec{y}_1 - \vec{y}_2\|_\infty \\ \alpha L < 1 &\Rightarrow \alpha < \frac{1}{L} \end{aligned}$$

Torej rešitev obstaja

Zvezna odvisnost od začetne točke

$$y_1(t, x_1) \quad y_2(t, x_2) ; \|x_1 - x_2\| < \delta \Rightarrow y_1(t, x_1) - y_2(t, x_2)$$

$$\|\mathcal{A}(t, y_1(t, x_1)) - \mathcal{A}(t, y_2(t, x_2))\|_\infty \leq$$

$$\leq \|x_1 - x_2\| + \left\| \int_{t_0}^t L \|y_1(t, x_1) - y_2(t, x_2)\| dt \right\|_\infty \leq$$

$$\leq \delta + \left\| \int_{t_0}^t L \|y_1 - y_2\|_\infty dt \right\|_\infty = \delta + L \|y_1 - y_2\|_\infty \cdot \alpha$$

$$\|y_1 - y_2\|_\infty (1 - L\alpha) \leq \delta$$

$$\|y_1 - y_2\|_\infty \leq \frac{\delta}{1 - L\alpha}$$

Tok vektorskega polja

$$\dot{\Phi}_t(x) = F(t, \Phi_t) \quad / \quad \frac{d}{dx}$$

$$(D_x \dot{\Phi}_t)(h) = (D_x F)(t, \Phi_t) \cdot (D_x \Phi_t)(h)$$

$$\leadsto \frac{d}{dt} (D_x \Phi) = D_x F(t, \Phi_t) \cdot D_x \Phi_{t_0} \quad D_x \Phi_{t_0} = I$$

Pomembni lastnosti:

$$1) \Phi_{t_0, t_0}(\vec{x}) = \vec{x}$$

$$2) \Phi_{t, s} \circ \Phi_{s, t_0} = \Phi_{t, t_0}$$

$$\text{Autonomni sistem} \leadsto \Phi_{t, t_0} = \Phi_{t+\tau, t_0+\tau} \Rightarrow \Phi_{t, t_0} = \Phi_{s, 0}$$

$$\leadsto 1) \Phi_0 = I$$

$$2) \Phi_{s+t} = \Phi_s \circ \Phi_t$$

$$\Rightarrow \Phi_t^{-1} = \Phi_{-t} \Rightarrow \Phi_t \text{ je difeomorfizem}$$

Izrek: Liouvilleova enačba

Φ_t tok avtonomnega polja $F(\vec{x})$ $J\Phi = \det(D\Phi)$

Potem $\frac{d}{dt} d\Phi_t(\vec{x}) = \operatorname{div}(F(\Phi_t(\vec{x}))) d\Phi_t(\vec{x})$

Dokaz: (za $n=2$)

Oznacimo $\Phi_t \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a(t, x, y) \\ b(t, x, y) \end{bmatrix}$

$$\dot{\Phi}_t \begin{bmatrix} x \\ y \end{bmatrix} = F(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix}) \leadsto \begin{bmatrix} \dot{a} \\ \dot{b} \end{bmatrix} = F(\begin{bmatrix} a \\ b \end{bmatrix}) / D_x$$

$$D_{\vec{x}}(\partial_t \begin{bmatrix} a \\ b \end{bmatrix}) = D_{\vec{x}} \left(\begin{bmatrix} f(a, b) \\ g(a, b) \end{bmatrix} \right)$$

$$\parallel$$

$$(D_{\vec{x}} \begin{bmatrix} a \\ b \end{bmatrix}) = \begin{bmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{bmatrix}$$

$$\begin{aligned} J &= \partial_t \det(D_x(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix})) = \partial_t \det \begin{bmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{bmatrix} = \\ &= \partial_t (\partial_x a \cdot \partial_y b - \partial_y a \cdot \partial_x b) = \\ &= \partial_t \partial_x a \cdot \partial_y b + \partial_x a \partial_t \partial_y b - (\partial_t \partial_y a \cdot \partial_x b + \partial_y a \cdot \partial_t \partial_x b) = \\ &= \partial_x \partial_t a \cdot \partial_y b - \partial_y \partial_t a \cdot \partial_x b + \partial_x a \cdot \partial_y \partial_t b - \partial_y a \cdot \partial_x \partial_t b = \\ &= \begin{vmatrix} \partial_x \dot{a} & \partial_y \dot{a} \\ \partial_x b & \partial_y b \end{vmatrix} + \begin{vmatrix} \partial_x a & \partial_y a \\ \partial_x \dot{b} & \partial_y \dot{b} \end{vmatrix} \end{aligned}$$

$$\text{ker velja } \begin{bmatrix} \dot{a} \\ \dot{b} \end{bmatrix} = F(\begin{bmatrix} a \\ b \end{bmatrix}) = \begin{bmatrix} f(a, b) \\ g(a, b) \end{bmatrix}$$

$$\text{je } \dot{a} = f(a, b) \leadsto \partial_x \dot{a} = f_a \partial_x a + f_b \partial_x b$$

$$\begin{aligned} J &= \begin{vmatrix} f_a \partial_x a + f_b \partial_x b & f_a \partial_y a + f_b \partial_y b \\ \partial_x b & \partial_y b \end{vmatrix} + \\ &+ \begin{vmatrix} \partial_x a & \partial_y a \\ g_a \partial_x a + g_b \partial_x b & g_a \partial_y a + g_b \partial_y b \end{vmatrix} = \end{aligned}$$

$$\begin{aligned} &= f_a (\partial_x a \cdot \partial_y b - \partial_y a \partial_x b) + f_b (\partial_x b \partial_y b - \partial_y b \partial_x b) \\ &+ g_a (\partial_x a \partial_y a - \partial_y a \partial_x a) + g_b (\partial_x a \partial_y b - \partial_y a \partial_x b) = \\ &= \begin{bmatrix} f_a \\ g_b \end{bmatrix} \cdot \begin{vmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{vmatrix} = \operatorname{div} F(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix}) \cdot \det(D_x \Phi_t \begin{bmatrix} x \\ y \end{bmatrix}) \end{aligned}$$

Trditev: fundamentalna matrika Φ je obrnljiva. ($\dot{\Phi} = A(t)\Phi$)

Dokaz: Recimo da $\Phi_{t_0, t_1} \vec{a} = \vec{0}$

$$\vec{y}(t) = \Phi_{t_0, t} \vec{a} = \sum_i \vec{a}_i \vec{X}_i(t)$$

$$\vec{y}(t_1) = \vec{0}$$

$\vec{y}$ z začetnim pogojem $\vec{y}(t_1) = 0$

reši enačbo $\dot{y} = F(t, y)$, $y(t_1) = 0$

to reši tudi $z \equiv 0$, torej po eksistenčnem izreku je $y \equiv 0$

torej $\vec{a} \equiv 0$ torej ni netrivialnega jedra.



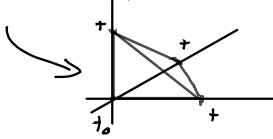
Izrek: Dysonova vrsta

• $A: [t_0, t] \rightarrow \text{Mat}_{n \times n}$ omejena po operatorski normi (norme, kjer $\|I\| = 1$)

Potem je fundamentalna matrika enačbe $\dot{\vec{x}} = A(t)\vec{x}$

$$\Phi_{t, t_0} = I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1) A(\tau_2) \dots A(\tau_n) d\tau$$

kjer je $\Delta_n(t) = \{(\tau_1, \dots, \tau_n) \in \mathbb{R}^n; t_0 \leq \tau_1 \leq \dots \leq \tau_n \leq t\}$



Dokaz: Pogledajmo kdaj velja $\dot{\Phi} = A(t)\Phi$ oziroma

$$\Phi = I + \int_{t_0}^t A(\tau) \Phi(\tau) d\tau$$

Rekurzivno kliče enačba sama sebe:

$$\begin{aligned} \Phi &= I + \int_{t_0}^t A(\tau_1) \left(I + \int_{t_0}^{\tau_1} A(\tau_2) \left(I + \int_{t_0}^{\tau_2} A(\tau_3) (I + \dots) d\tau_3 \right) d\tau_2 \right) d\tau_1 \\ &= I + \int_{t_0}^t A(\tau_1) d\tau_1 + \int_{t_0}^t d\tau_1 \int_{t_0}^{\tau_1} A(\tau_2) d\tau_2 + \int_{t_0}^t d\tau_1 \int_{t_0}^{\tau_1} d\tau_2 \int_{t_0}^{\tau_2} A(\tau_3) d\tau_3 \dots \\ &= I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1) \dots A(\tau_n) d\tau_1 \dots d\tau_n = \text{vrsta} \end{aligned}$$

To je izpeljava formule.

Preverimo ali je res rešitev

$$\begin{aligned} \dot{\Phi}_t &= \partial_t \sum_{n=1}^{\infty} \int_{\Delta_n(t)} \dots = \partial_t \sum_{n=1}^{\infty} \int_{t_0}^t d\tau_1 \int_{t_0}^{\tau_1} d\tau_2 \dots A(\tau_1) \dots A(\tau_n) d\tau_n \\ &\quad \partial_t \left(\int_{t_0}^t f(\tau) d\tau \right) = f(t) \\ &= \sum_{n=1}^{\infty} \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} \dots A(t) A(\tau_2) \dots A(\tau_n) d\tau_n = \\ &= A(t) \left(I + \sum_{n=2}^{\infty} \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} \dots A(\tau_2) \dots A(\tau_n) d\tau_n \right) = \\ &\stackrel{\uparrow}{=} A(t) \left(I + \sum_{n=1}^{\infty} \int_{t_0}^t d\tau_1 \int_{t_0}^{\tau_1} \dots A(\tau_1) \dots A(\tau_n) d\tau_n \right) \\ &\text{preimenujemo spremenljivke in v vrsti predstavimo n-1} \\ &= A(t) \cdot \Phi_t \end{aligned}$$

Preveriti moramo, ali lahko zamenjamo ∂_t in $\sum$ in ali vrsta Φ_t sploh obstaja.

$$\begin{aligned} \|\Phi_t\| &\leq 1 + \sum_{n=1}^{\infty} \int_{t_0}^t d\tau_1 \int_{t_0}^{\tau_1} \dots \|A(\tau_1)\| \|A(\tau_2)\| \dots \|A(\tau_n)\| d\tau_n \leq \\ &\leq 1 + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} M^n = 1 + \sum_{n=1}^{\infty} \frac{(t-t_0)^n}{n!} M^n = \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} ((t-t_0)M)^n = e^{(t-t_0)M} < \infty \end{aligned}$$

Torej Φ_t obstaja.

Ali lahko zamenjamo ∂_t in $\sum$ v dokazu?

$$\sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1) \dots A(\tau_n) d\tau \leq \sum_{n=1}^{\infty} \frac{1}{n!} ((t-t_0)M)^n < \infty$$

Torej po Weierstrassovem M-testu je to enakomerna konvergenca, torej $\partial_t \sum \dots = \sum \partial_t \dots$

Izvedi: Formula, ko A_j : komutirajo

Če velja $\forall t_1, t_2 \in [0, b]$

$$[A(t_1), A(t_2)] = A(t_1)A(t_2) - A(t_2)A(t_1) = 0$$

Potem za $\dot{\Phi}_+ = A(t)\Phi_+$ $\Phi_0 = I$ velja rezultat

$$\Phi_+ = e^{\int_0^+ A(\tau) d\tau}$$

Dokaz:

$$\Phi_+ = I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1) \dots A(\tau_n) d\tau$$

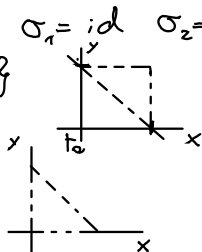
Definirajmo

$$\Delta_n^{\sigma}(t) = \{(\tau_1, \dots, \tau_n) \in \mathbb{R}^n; t_0 < \tau_{\sigma(1)} < \dots < \tau_{\sigma(n)} < t\}$$

za $n=2$ imamo dve permutaciji: $\sigma_1 = \text{id}$ $\sigma_2 = (1\ 2)$

$$\Delta_n^{\sigma_1}(t) = \{(x, y); t_0 < x < y < t\}$$

$$\Delta_n^{\sigma_2}(t) = \{(x, y); t_0 < y < x < t\}$$



$$\Delta_n^{\sigma_1}(t) \perp \Delta_n^{\sigma_2}(t) = \square_2(t)$$

$$\square_n(t) = \{(\tau_1, \dots, \tau_n) \in \mathbb{R}^n; \forall i, j, i \neq j \Rightarrow \tau_i \neq \tau_j\}$$

↗ brez diagonal

$$\text{Podobno } \bigsqcup_{\sigma \in S_n} \Delta_n^{\sigma}(t) = \square_n(t)$$

Ker je $\Pi A(\tau_i) = \Pi A(\tau_{\sigma(i)})$ (ker matrice komutirajo)

$$\text{velja } \int_{\Delta_n^{\sigma}} A(\tau_{\sigma(1)}) \dots A(\tau_{\sigma(n)}) d\tau = \int_{\Delta_n} A(\tau_1) \dots A(\tau_n) d\tau$$

$$\Rightarrow \int_{\square_n(t)} A(\tau_1) \dots A(\tau_n) d\tau = \sum_{\sigma \in S_n} \int_{\Delta_n^{\sigma}(t)} A(\tau_1) \dots A(\tau_n) d\tau =$$

$$= n! \int_{\square_n(t)} A(\tau_1) \dots A(\tau_n) d\tau$$

$$\Rightarrow \Phi_+ = I + \sum_{n=1}^{\infty} \frac{1}{n!} \int_0^+ A(\tau_1) d\tau_1 \int_0^+ A(\tau_2) d\tau_2 \dots \int_0^+ A(\tau_n) d\tau_n =$$

$$= I + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\int_0^+ A(\tau) d\tau \right)^n = e^{\int_0^+ A(\tau) d\tau}$$

Izrek: Globalnost rešitev

Naj bo $A(t) \leq M$ za $\forall t \in [t_0, t_1]$

Potem je rešitev $\vec{x}(t)$ za $\ddot{\vec{x}} = A(t)\vec{x}$ $\vec{x}(0) = \vec{a}$
definirana na celém intervalu $[t_0, t_1]$

Dokaz:

$\vec{x}(t) = \Phi_{t_0, t} \vec{a}$ Ali: Φ_t dostaja povsod?

$$\|\Phi_{t_0, t}\| = \left\| I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1) \dots A(\tau_n) d\tau \right\| \leq$$

$$\leq 1 + \sum_{n=1}^{\infty} (t-t_0)^n \cdot \frac{1}{n!} M^n = e^{(t-t_0)M} \leq e^{(t_1-t_0)M} < \infty$$



Izrek: Liouvilleova formula

$$\text{Naj bo } \dot{\Phi}_t = A(t) \cdot \Phi_t$$

$$\text{Potem velja } \frac{d}{dt} \det(\Phi_t) = \text{sl}(A(t)) \cdot \det \Phi_t$$

$$\text{Dokaz: } \Phi_t = \begin{bmatrix} x_{11}(t) & \dots & x_{1n}(t) \\ \vdots & & \vdots \\ x_{n1}(t) & \dots & x_{nn}(t) \end{bmatrix}$$

$$\det(\Phi_t) = \sum_{\sigma \in S_n} (-1)^{\text{sgn}(\sigma)} x_{1\sigma(1)} \dots x_{n\sigma(n)}$$

$$\frac{d}{dt} \det(\Phi_t) = \sum_{\sigma \in S_n} (-1)^{\text{sgn}(\sigma)} \sum_{i=1}^n \dot{x}_{i\sigma(i)} x_{1\sigma(1)} \dots x_{n\sigma(n)} =$$

$$\dot{\Phi} = A(t) \Phi \Rightarrow \dot{x}_{i\sigma(i)} = \sum_{j=1}^n a_{ij} x_{j\sigma(i)}$$

$$= \sum_{\sigma \in S_n} (-1)^{\text{sgn} \sigma} \cdot \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_{1\sigma(1)} \dots x_{j\sigma(i)} \dots x_{n\sigma(n)} =$$

$$= \sum_{i=1}^n \sum_{j=1}^n a_{ij} \sum_{\sigma \in S_n} (-1)^{\text{sgn} \sigma} x_{1\sigma(1)} \dots x_{j\sigma(i)} \dots x_{n\sigma(n)}$$

imamo σ_1 in $\sigma_2 = (i j) \sigma_1$ za $i \neq j$

$$\begin{aligned} \text{Potem je } & (-1)^{\text{sgn} \sigma_1} x_{1\sigma_1(1)} \dots x_{j\sigma_1(i)} \dots x_{i\sigma_1(i)} \dots x_{n\sigma_1(n)} = \\ & = - (-1)^{\text{sgn} \sigma_2} x_{1\sigma_2(1)} \dots x_{j\sigma_2(j)} \dots x_{i\sigma_2(j)} \dots x_{n\sigma_2(n)} \end{aligned}$$

Torej vsi členi kjer $i \neq j$ se pokrajšajo

$$= \sum_{i=1}^n a_{ii} \sum_{\sigma \in S_n} (-1)^{\text{sgn} \sigma} x_{1\sigma(1)} \dots x_{n\sigma(n)} = \sum_{i=1}^n a_{ii} \det \Phi_t =$$

$$= \text{sl}(A(t)) \det \Phi_t$$

Trditve: A, B diagonalizabilni in $AB=BA$
Potem obstaja prehodna matrika P , da sta
 PAP^{-1} in PBP^{-1} diagonalni.

Dokaz:

$$Av = \lambda v$$

$$ABv = BAv = B(\lambda v) = \lambda Bv$$

$\Rightarrow Bv$ je lastni vektor za A

$E_\lambda = \{v \in \mathbb{R}^n; Av = \lambda v\}$... prestor lastnih
vektorjev za lastno vrednost λ

$$v \in E_\lambda \Rightarrow Bv \in E_\lambda$$

Torej B lahko gledamo kot $B_\lambda : E_\lambda \rightarrow E_\lambda$

B_λ lahko diagonaliziramo, ker lahko diagonaliziramo B

$$P_\lambda B_\lambda P_\lambda^{-1} = D_{B_\lambda} \quad \text{v bazi } \{\vec{v}_\lambda^1, \dots, \vec{v}_\lambda^n\}$$

$$Av_\lambda^i = \lambda v_\lambda^i \Rightarrow P_\lambda A P_\lambda^{-1} = \begin{bmatrix} \lambda & & \\ & \ddots & \\ & & \lambda \end{bmatrix}$$

Tako naredimo za $\forall \lambda$ in dobimo

$$P = \begin{bmatrix} P_1 & & \\ & \ddots & \\ & & P_k \end{bmatrix}$$

da je $PAP^{-1} = D_A$
in $PBP^{-1} = D_B$